Reg. No.											

MANIPAL INSTITUTE OF TECHNOLOGY

(A constituent unit of MAHE, Manipal 576104)

III SEM B.Tech (BME) DEGREE MAKEUP EXAMINATIONS, DEC/JAN 2019-20

SUBJECT: DIGITAL ELECTRONICS (BME 2153)
(REVISED CREDIT SYSTEM)
Thursday, 26th December, 2019: 8.30 am to 11.30 am.

TIME: 3 HOURS **MAX. MARKS: 50**

Instructions to Candidates:

1. Answer ALL questions.
2. Draw labeled diagrams wherever necessary.

1A. Depicting the necessary steps and calculations perform the following: 04

(a)*Subtract the following BCD numbers using* $2's$ *complement method*

(i) 36 *from* 75 (ii) 98 *from* 72

(b) *Convert into,*

$(i)\ (45.25)_{10} = (\quad)_2$ $(ii)\ (324.32)_8 = (\quad)_{10}$

1B. Simplify the following functions to a minimum number of literals using Boolean simplifications. 03

$(i)\ f(xyz) = (\bar{x} + xy\bar{z})(x + \bar{x}\bar{y}z) + (\bar{x} + xy\bar{z})$

$(ii)\ f(xyz) = \bar{x}\bar{y}z + y\bar{z} + \bar{x}yz + xyz$

$(iii)\ f(xyz) = \overline{[\bar{x}(\bar{y} + \bar{z})(x + y + \bar{z})]}$

1C. Realize a half adder and draw the circuit using only five numbers of two input NOR gates. 03

2A. Design and draw the circuit of 1 digit BCD adder. Explain its operation with suitable examples. 04

2B Simplify the following function using Karnaugh map and realize using only NAND gates. 03

$F(ABCD) = \sum m(2,3,4,5,6,7,12,13$

2C. Realize and draw the circuit of 2×4 line decoder using appropriate gates with active low outputs and active low Enable line. 03

3A. Design and draw the decoder circuit for seven segment common cathode display for displaying the numbers 0 to 7. 04

3B. Realize and draw the circuit of Octal to Binary Encoder using diode matrix. 03

3C Design and draw 8×1 Multiplexer to realize the following function. 03

$F(A,B,C,D) = \sum m(0,1,3,5,6,7,8,9,11,12,15$

4A. Realize and draw the circuit of a ROM to obtain the squares of Octal numbers (0 to7) 04

4B. Draw the circuit of a Master slave JK flip flop using NAND gates. Explain how the race around condition is eliminated in this flip flop? 03

4C Draw the circuit of an asynchronous Decade counter using Toggle flip flops. Write the count sequence and draw the timing waveforms. 03

5A. Design a synchronous counter to count the following count sequence using JK flip flops. Also verify the counting with timing waveforms. 04

$000, 001, 011, 100, 101, 111, 000, \ldots \ldots$

5B. Draw the circuit of a 4 bit shift register using D flip flops to operate as (i) SIPO (ii) SISO. Explain the operation with the appropriate truth table. 03

5C. Draw the circuit of a Mod-8 Johnson counter. List the legal and illegal count sequences. Give a remedy circuit to avoid the illegal counts. Also draw the necessary decoder circuit to read the legal counts. 03